

THE VIEWER - Visual Changelog

A functioning data-flow panel for every version (chronological). Dark (R3) - PDF (R5) - accompanies docs/CHANGELOG.md (R4).

feature fix rule/standard

0.1.0

Architecture & diagrams

Locked: local web app + SQLite/FTS5 + text-first. Established rules R1 (backwards-compatible) and R2 (diagram per addition).



0.2.0

Indexing engine + OCR pipeline

viewer_ingest + migration 0001 (documents/pages/pages_fts/jobs/runs). Text-first, resumable, idempotent; durable OCR job queue.



0.3.0

Onboarding + 104th parts request

Migration 0002 (sessions/faults/request_items). Offline app + dark UI; clean PDF replica of the 104th ECC sheet, header from the modal.



0.4.0

Full-corpus extraction + durability

Time-boxed resumable batches + cheap skip (size+mtime). Fixed corruption by switching to EXCLUSIVE + on-disk TRUNCATE Journal.



0.5.0

PyMuPDF text backend

Text extraction moved to PyMuPDF (pip) with Poppler fallback - faster and removes the Poppler hard dependency.



0.6.0

RapidOCR engine; full run

OCR engine = RapidOCR (pip, no admin). Full crawl COMPLETED on PC: 40,291 docs, ~1.87M searchable text pages, ~133k queued for OCR.



0.6.1

OCR detail + example

Detailed OCR diagram + a real before/after example page (recovered part numbers from a scanned parts page).



0.7.0

Dark diagram standard (R3)

Rule R3: all diagrams dark-themed + a PDF version. Converted existing diagrams; viewer.html switched to dark.



0.8.0

Search GUI: document viewer

On-demand page rendering (PyMuPDF) + viewer pane: click a result to see the real manual page, navigate, add the part to the request.



0.8.1

OCR reliability fixes

Root cause: orphan sandbox-path rows failed instantly. Added cleanup + switched to thread-based OCR. Verified real text, zero failures.



0.9.0

Forked: GPU + Lite

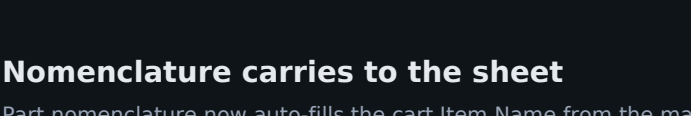
One codebase, two profiles: GPU/production (priority) and Lite/portable (self-contained folder + finished index). Flags --gpu/-dpl.



0.9.1

Changelog standard (R4)

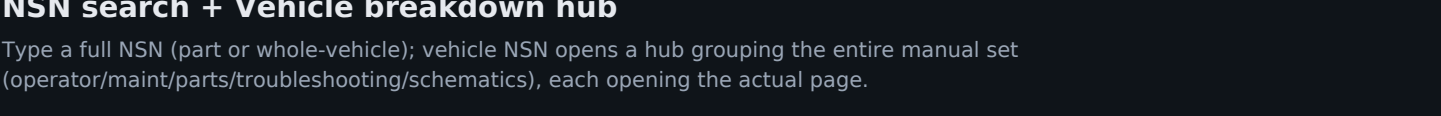
Rule R4: a versioned changelog entry accompanies every change. Retroactive CHANGELOG.md created.



0.9.2

Visual changelog (R5)

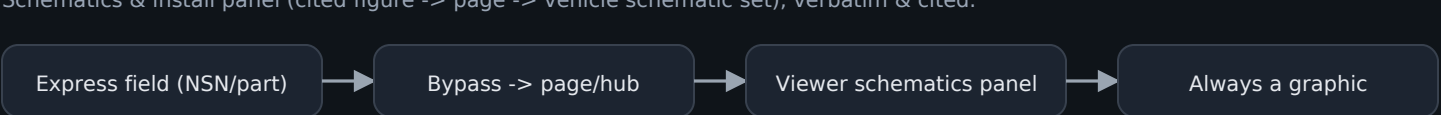
Rule R5: every changelog entry also gets a detailed graphical explanation + a functioning diagram PDF (this document).



0.9.3

Launcher pip self-upgrade

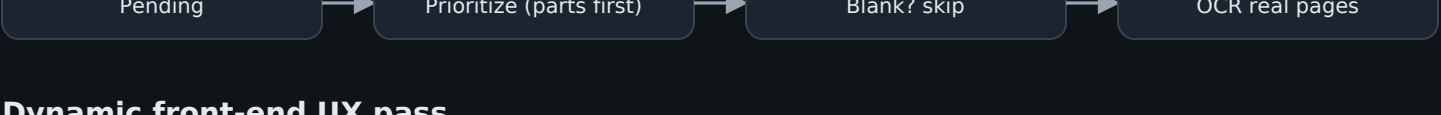
All setup/launch files now run pip install --upgrade pip before installing packages, so nothing is missing and the app launches cleanly.



0.9.4

Nomenclature carries to the sheet

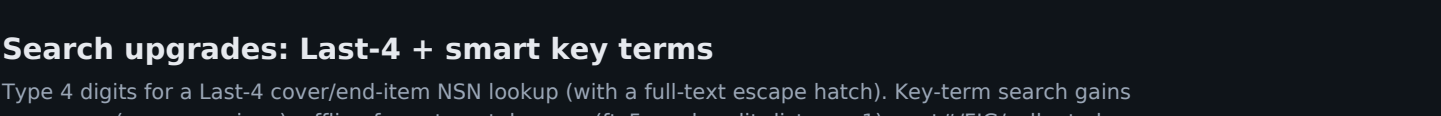
Part nomenclature now auto-fills the cart item Name from the matched manual line (or search term), so it carries onto the parts request sheet.



0.10.0

NSN search + Vehicle breakdown hub

Type a full NSN (part or whole-vehicle); vehicle NSN opens a hub grouping the entire manual set (operator/maint/parts/troubleshooting/schematics), each opening the actual page.



0.11.0

Express modal + Schematics on every page

Modal 'What do you need?' jumps straight to the page/hub on exact NSN/part#: the viewer always shows a Schematics + install panel (cited figure-> page-> vehicle schematic set), verbatim + cited.



0.12.0

OCR speedup: skip-junk + prioritize

Skip-the-junk: blank pages skipped with no OCR inference (~0.01s). Prioritized queue OCRs parts catalogs first. Migration 0003 + narrowed FTS trigger (faster queues).



0.13.0

Dynamic front-end UX pass

Home screen browses every vehicle + recents (/api/vehicles, /api/sessions). Results gain filters/counts by vehicle/type/source. Viewer adds zoom, a thumbnail strip, and highlight-the-hit (/page?hl via PyMuPDF



0.14.0

Search upgrades: Last-4 + smart key terms

Type 4 digits for a Last-4 cover/end-item NSN lookup (with a full-text escape hatch). Key-term search gains synonyms (synonyms.json), offline fuzzy typo tolerance (fts5vocab, edit-distance 1), part#/FIG/callout phrase



0.15.0

Modal aligned to the 104th sheet header

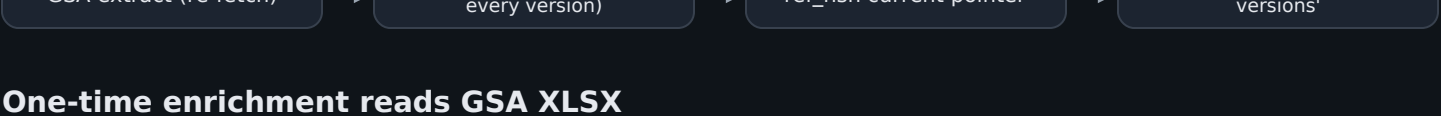
Onboarding modal now mirrors the 104th sheet header 1:1 (exact labels & order); header fields always shown (Simple only hides cart FEDLOG); a live paper-style header preview fills in as you type. Field IDs/SESSION keys



0.16.0

Self-learning search + calmer UI

Surfaces what you already log: GET /api/popular ranks parts that reached a generated sheet by frequency+recency. Drives a single rotating example, a 'commonly requested' quick-pick row, and popularity-ranked results



0.17.0

Tech Status from fault+part (PMCS-cited)

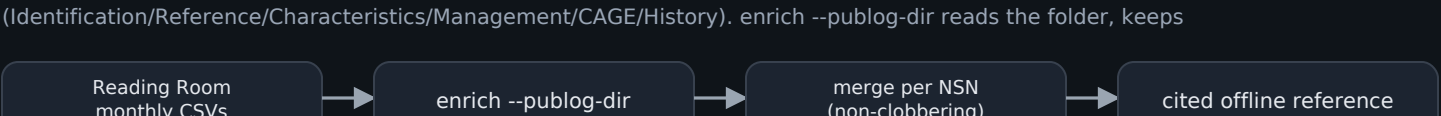
Suggests equipment status from the fault: searches the vehicle's PMCS 'Not Fully Mission Capable IF' criteria (cited, TM+page) -> deadline=NMCS; else prior history; else manual. Mandatory confirm gate at export (server



0.18.0

Switchable layouts behind Settings

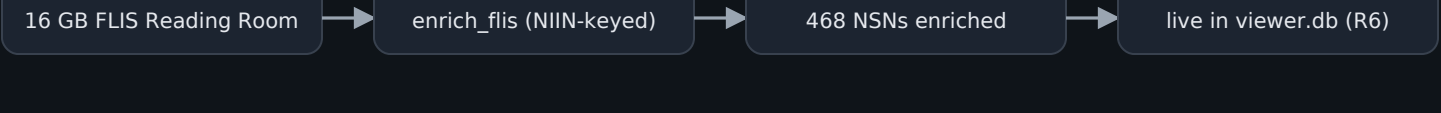
A Settings panel consolidates the toggles + 4 named presets (Simple/Advanced/Shopfloor/Compact), per-device persist + Reset; legacy keys migrated. Presentation only: a layout is just CSS classes + 2 defaults; the



0.19.0

Structured parts index (Phase 1) + coverage

Extracts a cited NSN->figure->page->vehicle index from RPSTL pages (migration 0004; 28,330 records on the sample). Cart shows the cited catalog figure + auto-fills FIG; vehicle hub shows '% searchable'. Multi-sheet



0.20.0

Online -> offline reference enrichment

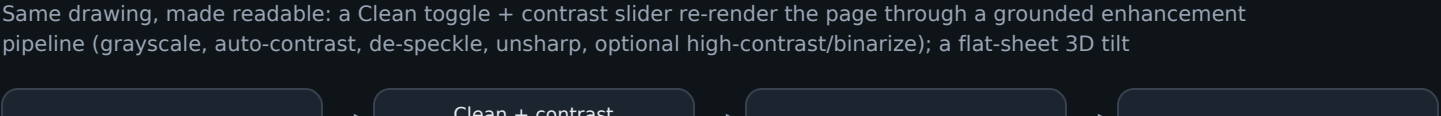
One-time online enrichment, offline after: public-domain FED-STD-H2B hardware/thread dims + official GSA NSN extract (filtered to in-index NSNs). Separate cited tables (migration 0005), labeled 'External reference';



0.21.0

Append-only NSN enrichment (rule R6)

Rule R6: always add, never remove info (even outdated). NSN enrichment is now append-only/versioned (migration 0006, ref_nsn_log) - old + new both kept; current = latest. Space/scope: NSNs are cheap text (tens of MB); whole



0.21.1

One-time enrichment reads GSA XLSX

enrich now reads the official GSA NSN Extract directly as .xlsx (the actual data.gov format) as well as csv. Engine stays offline; one-shot hand-run filler. Source is public-domain (CC0) but the GSA Advantage subset,



0.22.0

PUB LOG reference ingest (POC)

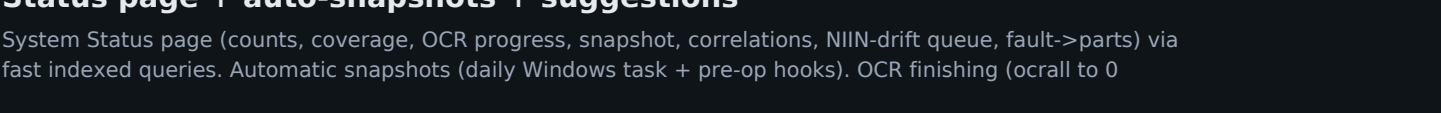
POC: enrich -publog ingests the authoritative DLA PUB LOG export - fills NSN->item name, part#/CAGE (MCRD), characteristics=size for Tier 2.5 (CHAR), AAC + substitutes (MDIsS). Migration 0007; append-only/cited (R6);



0.23.0

PUB LOG via FLIS Reading Room (direct CSVs)

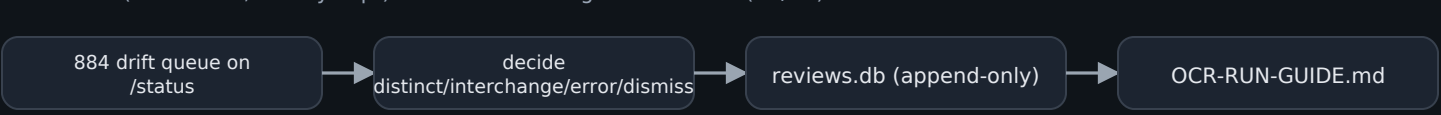
Correction: no PUB LOG Windows app needed - the FLIS Reading Room publishes monthly CSVs (Identification/Reference/Characteristics/Management/CAGE/History). enrich -publog-dir reads the folder, keeps



0.24.0

PUB LOG enrichment RUN on the live index

Run the real DLA FLIS catalog into the live index: 468 NSNs got item names (406), part numbers (463), AAC (451), decoded characteristics/dimensions (421), unit prices. enrich_flis() productizes the NIIN-keyed ingest



0.25.0

Overnight: full-catalog enrichment + 2D->3D + rollback

On the live index: 227,908 parts records (45,068 NSNs); FLIS enrichment of 41,701 NSNs (names, part#/CAGE, dimensions, AAC, price, vintage date, supersession cross-refs, multiple part choices). New: FLIS year tag +



0.26.0

3D viewer: family shapes + material/colour

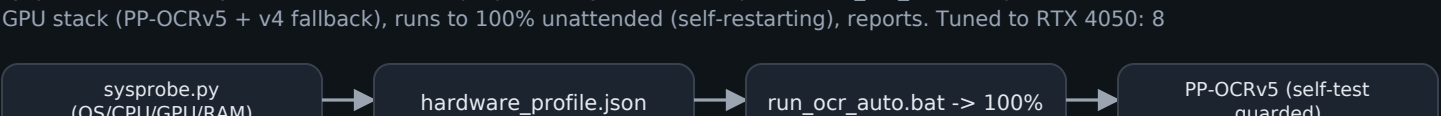
Representative 3D viewer upgraded: renders family shapes from the item name with depth-sorted filled facas; tints to the FLIS-stated colour (cited) or a material-based representative tint (steel>grey, aluminum>silver.



0.27.0

Schematic legibility viewer

Same drawing, made readable: a Clean toggle + contrast slider re-render the page through a grounded enhancement pipeline (grayscale, auto-contrast, de-speckle, unsharp, optional high-contrast/binarize); a flat-sheet 3D tilt



0.28.0

Hi-fi loupe + correlations + tests

Three additive wins: (1) the loupe re-rasterises just the page region under the cursor at high DPI (~21x more pixels - sharp at any zoom, nothing invented); (2) a deletable correlations sidebar connects links the flat



0.29.0

Data protection: safeguard + recovery

Root cause of the 'truncation' found: a sandbox read-cache artifact at the host->guest boundary - the Windows files were never damaged. Built real protection anyway: engine/safeguard.py does atomic writes, SHA-256



0.30.0

Schematic orientation + HD

Schematic viewing controls: dual-axis tilt (rotateY + rotateX, honest flat-sheet); a horizontal mirror to orient from the opposite side, with each word box (new /api/pagewords from PyMuPDF) re-drawn un-mirrored so labels stay



0.31.0

Status page + auto-snapshots + suggestions

System Status page (counts, coverage, OCR progress, snapshot, correlations, NIIN-drift queue, fault->parts) via fast indexed queries. Automatic snapshots (daily Windows task + pre-op hooks). OCR finishing (ocrall to 0



0.32.0

NIIN-review workflow + OCR run guide

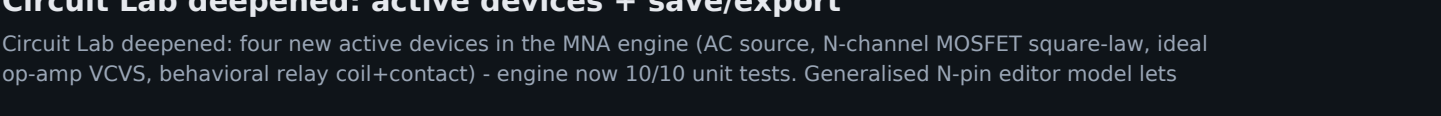
The 884 NIIN-drift findings become an actionable confirm/reject queue; decisions persist append-only in reviews.db (latest wins, history kept). Plus an OCR run guide. Additive (R1/R6).



0.33.0

NSN alias map + GPU readiness

Confirmed-interchangeable NSNs find each other in search (grounded, reversible). gpu_check.py GPU prioritized. Prioritized the live OCR queue; OCR 1.6%, 23/23 pillar tests.



0.34.0

Loupe, seamless loupe

Tight reworked: instant local magnification every frame, high-DPI crop sharpens on pause, cached, wheel-zoom. Cohesive control styling. Presentation-only (R1/R6).



0.35.0

Page zoom: slider + scroll-to-cursor

Continuous zoom slider + scroll-to-zoom centered on the cursor (loupe off); double-click reset; stacks into the CSS transform with tilt/mirror. Presentation-only (R1/R6).



0.36.0

Hardware probe + autonomous adaptive GPU OCR

sysprobe scans OS/Python/CPU/RAM/GPU/laptop+battery -> resource profile; run_ocr_auto.bat probes, installs the GPU stack (PP-OCRv5 + v4 fallback), runs to 100% unattended (self-restarting), reports. Tuned to RTX 4050: 8

0.37.0

COMPLETE backward compatibility: Win11 -> Vista

Complete feature compatibility Win11 down to Win7/Vista via per-OS engine substitution (PyMuPDF<->Poppler, RapidOCR<->Tesseract; stdlib core). Only GPU to Win7/Vista via per-OS engine substitution (PyMuPDF<->Poppler, RapidOCR<->Tesseract; stdlib core). Only GPU acceleration is Win10+ - a speed booster, not a feature.

0.38.0

Dual-track changelog scaffolding (R7)

Rule R7: legacy-builds get a dual-track changelog that branches at creation and shows backports. Dual-tracking: CHANGELOG-LEGACY.md starts at 0.37.0-legacy; branched-timeline generator -> CHANGELOG-DUALTRACK.pdf; parity line

0.39.0

3D Library + Schematics Library + Reset

Reset buttons everywhere (schematic viewer clears tilt/zoom/mirror; 3D viewer Reset + double-click). 3D Library (/3d): searchable gallery of 20,869 dimensioned parts, live mini 3D thumbnails. Schematics Library

0.40.0

Make it pop: WebGL 3D + dynamic schematics

Real-time WebGL 3D viewer (engine/ui/g13d.js, dependency-free, served at /g13d.js); glossy multi-light shading, antialiasing, smooth normals for round families, idle turntable, orbit/zoom/reset; the 3D Library uses it with

0.41.0

Schematics: explore from every angle + rail

The dedicated /schematics viewer now explores a sheet as thoroughly as possible: tilt X and tilt Y sliders (CSS perspective, honest flat-sheet) view it from the left, right, above, below and any angle between; a mirror flips

0.42.0

Circuit Lab: overlay editor + real-time simulator

Circuit Lab (/circuitlab): build or trace a circuit on top of a real schematic, then watch it run - a learning/advanced-display tool. Dependency-free MNA engine (engine/ui/circuitsim.js): G=1, backward-Euler

0.43.0

Look-Alike Parts recognizer

Look-Alike Parts (/partdiff, part_differences()) + /api/partdiff): tells apart parts that look identical in the manual but are functionally different. Finds every catalogued part sharing a name, collapses to distinct NSNs.

0.44.0

Circuit Lab deepened: active devices in the MNA engine (AVE export)

Circuit Lab deepened: four new device devices in the MNA engine (AVE source, N-channel MOSFET square-law, ideal op-amp VCVS, behavioral relay coil +contact) - engine now 10/10 unit tests. Generalised N-pin editor model lets